Martín Hernández Salinas

Reinforcement Learning ▪ **Machine Learning** ▪ **Scientific Computing** ▪ **Optimal Control**

Reinforcement-learning and scientific-computing researcher with a Ph.D. in Mathematics (*summa cum laude*) and postdoctoral experience in Statistics & Data Science at UCLA. I develop mathematically grounded and computationally scalable methods for continuous-time and multi-agent learning, combining theoretical analysis with algorithm design, research-software implementation and numerical validation. My current work focuses on continuous-time and mean-field reinforcement learning, deep neural networks, randomized algorithms, and optimal control.

martinh@g.ucla.edu • +1 310 254 9741 • Los Angeles, USA
martinshs.github.io • github.com/Martinshs • linkedin.com/in/mahernandezs • ORCID: 0009-0003-4799-6853

ACADEMIC APPOINTMENTS

Postdoctoral Researcher 11/2025 – Present
Department of Statistics & Data Science, UCLA, USA
Advisor: Prof. Yuhua Zhu. Research on continuous-time reinforcement learning and mean-field control, including model-free policy-gradient and actor–critic methods from discrete-time observations; mathematical error and convergence analysis; and computational validation of the proposed algorithms.

Doctoral Researcher 09/2021 – 09/2025
Department of Mathematics, FAU Erlangen–Nürnberg, Germany
Supervisor: Prof. Enrique Zuazua. Research at the intersection of machine learning, optimal control, numerical analysis, and scientific computing, including continuous-time dropout for neural ODEs, constructive deep-ReLU approximation, random-batch methods, PDE control, and randomized domain decomposition.

EDUCATION

Ph.D. in Mathematics 09/2021 – 09/2025
FAU Erlangen–Nürnberg, Germany
Thesis: *From Optimal Control to Random and Neural Network Approximation.* Supervisor: Prof. Enrique Zuazua. Awarded *summa cum laude*.

M.Sc. in Mathematics 03/2019 – 03/2021
Universidad Técnica Federico Santa María (UTFSM), Chile
Graduated with the highest distinction. Supervisors: Prof. Rodrigo Lecaros and Prof. Sebastián Zamorano.

Professional Degree in Mathematical Engineering 03/2015 – 03/2021
Universidad Técnica Federico Santa María (UTFSM), Chile
Graduated with the highest distinction. Supervisors: Prof. Rodrigo Lecaros and Prof. Sebastián Zamorano.

RESEARCH AREAS & COMPUTATIONAL EXPERTISE

- **Reinforcement learning and control:** continuous-time RL, policy-gradient and actor–critic methods, mean-field control, stochastic and deterministic optimal control.
- **Machine learning:** deep ReLU networks, neural ODEs, continuous-time dropout, approximation and memorization, learning and optimization algorithms.
- **Randomized and numerical methods:** random-batch and mini-batch methods, interacting systems, PDE-constrained optimization, domain decomposition, and finite-element computation.
- **Programming and research software:** Python, PyTorch, TensorFlow, NumPy, SciPy, Matplotlib, MATLAB, FEniCS/DOLFIN, Git; algorithm implementation, numerical benchmarking, and reproducible computational workflows.

PEER-REVIEWED PUBLICATIONS

[1] **Random Batch Methods for PDE Control on Graphs**
Martin Hernandez and Enrique Zuazua
Accepted for publication in *Journal of Computational Physics* (JCP) (2026). arXiv:2506.11809

[2] **Random Domain Decomposition for Parabolic PDEs**
Martin Hernandez
Accepted for publication in *SIAM Journal on Numerical Analysis* (SINUM) (2026). arXiv:2508.21557

[3] **Turnpike phenomenon for averaged optimal control**
Martin Hernandez, Martin Lazar, and Sebastián Zamorano
Optimization (2026), 1–24. DOI:10.1080/02331934.2026.2615776

[4] **Mini-batch Descent in Semiflows**
Alberto Domínguez Corella and **Martin Hernandez**
ESAIM: Control, Optimisation and Calculus of Variations **31** (2025), Article 28. DOI:10.1051/cocv/2025018

[5] **Uniform Turnpike Property and Singular Limits**
Martin Hernandez and Enrique Zuazua
Acta Applicandae Mathematicae **190** (2024), Paper No. 3, 33 pp. DOI:10.1007/s10440-024-00640-7

[6] **Averaged Turnpike Property for Differential Equations with Random Constant Coefficients**
Martin Hernandez, Rodrigo Lecaros, and Sebastián Zamorano
Mathematical Control and Related Fields **13**(2) (2023), 808–832. DOI:10.3934/mcrf.2022016

FORTHCOMING BOOK CHAPTER

[1] **Domain Decomposition for Gas Network Control**
Falk Hante, **Martin Hernandez**, Günter Leugering, Martin Schmidt, and Enrique Zuazua
In *Mathematical Modelling, Simulation and Optimization Using the Example of Gas Networks*, International Series of Numerical Mathematics, Vol. 174, Birkhäuser/Springer, pp. 361–410; forthcoming February 2027. ISBN:978-3-032-34969-9

PREPRINTS

[1] **Mean-Field PhiBE: Continuous-Time Mean-Field Reinforcement Learning from Discrete-Time Data**
Erhan Bayraktar, **Martin Hernandez**, Qinxin Yan, and Yuhua Zhu
Preprint (2026). arXiv:2606.26498

[2] **Policy Gradient for Continuous-Time Mean-Field Control**
Erhan Bayraktar, **Martin Hernandez**, Qinxin Yan, and Yuhua Zhu
Preprint (2026). arXiv:2605.20718

[3] **Convergence, Design and Training of Continuous-Time Dropout as a Random Batch Method**
Antonio Àlvarez-López and **Martin Hernandez**
Preprint (2025). arXiv:2510.13134

[4] **Constructive Universal Approximation and Finite Sample Memorization by Narrow Deep ReLU Networks**
Martin Hernandez and Enrique Zuazua
Preprint (2025). arXiv:2409.06555

FELLOWSHIPS & RESEARCH SUPPORT

- **2022–2026:** ANID–DAAD Bilateral Abroad Scholarship (*Acuerdo Bilateral en el Extranjero ANID–DAAD*), competitive doctoral fellowship.
- **2023–2025:** DAAD–CAPES Programs for Project-Related Personal, grant 57703041, *Control and numerical analysis of complex systems.*
- **2022–2026:** Research support through SFB/TRR 154, *Mathematical Modelling, Simulation and Optimization Using the Example of Gas Networks.*
- **2021–2022:** Doctoral research support through Deutsche Forschungsgemeinschaft (DFG), Project ID 239904186 – SFB/TRR 154.
- **2019–2021:** Graduate scholarship, Dirección de Postgrados y Programas (DPP), Universidad Técnica Federico Santa María.

TEACHING EXPERIENCE

Summer School Lecturer 2025
Universidade Federal Fluminense (UFF), Brazil
Delivered lectures on *Introduction to Long-Time Optimal Control*.

Summer School Lecturer – Introduction to PyTorch 2024
ML&OC Summer School, Gaeta, Italy; EECI-IGSC M18 Summer School, Dubrovnik, Croatia
Designed and delivered international hands-on lectures covering automatic differentiation, optimization, regression, multilayer neural networks, ResNets, and classification with PyTorch; developed the complete open teaching material.

Part-Time Lecturer in Mathematics 2021
Department of Mathematics, UTFSM, Chile
Taught undergraduate Differential Calculus (2021-2) and Integral Calculus (2021-1) to cohorts of approximately 60 students; delivered two weekly class sessions and prepared course materials.

Undergraduate Teaching Assistant 2018 – 2020
Department of Mathematics, UTFSM, Chile
Led weekly problem-solving sessions and graded in-course examinations for cohorts of approximately 60 students in differential and integral calculus, single- and multivariable calculus, and algebra and geometry.

SELECTED RESEARCH SOFTWARE & TEACHING MATERIALS

- **Introduction to PyTorch:** designed an open, hands-on tutorial on automatic differentiation, regression, optimization, MLPs, ResNets, MNIST, and mini-batch training.
- **Random Minimizing Movement for PDEs:** developed FEniCS solvers and reproducible 1D/2D numerical experiments associated with *Mini-batch Descent in Semiflows*.
- **Continuous-Time Dropout as a Random Batch Method:** co-developed the research code for the numerical experiments and training algorithms associated with *Convergence, Design and Training of Continuous-Time Dropout as a Random Batch Method*.
- **Constructive Deep-ReLU Classification:** implemented an algorithm that constructs network parameters directly from labeled data without numerical training.
- **PINNs for the Heat Equation:** co-developed the PyTorch/deepxde implementation and verified the mathematical formulation for a parabolic PDE solver.

TALKS & LECTURES

- **07/2026** – Talk, *Dynamics, Deep Learning, and Control in Nonlinear PDEs*, Ettore Majorana Foundation and Centre for Scientific Culture, Erice, Italy.
- **05/2026** – Talk, *Multi-Agent and Mean Field Reinforcement Learning*, Chicago, USA.
- **04/2026** – Talk, Southern California Applied Mathematics Symposium, Los Angeles, USA.
- **01/2025** – Research talk, Technical University of Darmstadt, Germany; scientific visit to Prof. Jan Giesselmann.
- **01/2025** – Lectures, Universidade Federal Fluminense, Brazil; scientific visit to Prof. Juan Límaco.
- **12/2024** – Research talk, Humboldt University of Berlin, Germany; scientific visit to Prof. Falk Hante.
- **26/08/2024** – Talk, *X Partial Differential Equations, Optimal Design and Numerics*, Benasque, Spain.
- **23/08/2024** – Talk, *X Partial Differential Equations, Optimal Design and Numerics*, Benasque, Spain.
- **19/08/2024** – Talk, *X Partial Differential Equations, Optimal Design and Numerics*, Benasque, Spain.
- **06/2024** – Lecturer, EECI-IGSC 2024: Control and Machine Learning (M18), Dubrovnik, Croatia.
- **05/2024** – Lecturer, ML&OC24: Machine Learning and Optimal Control Summer School, Gaeta, Italy.
- **08/2023** – Talk, 10th International Congress on Industrial and Applied Mathematics (ICIAM), Tokyo, Japan.
- **02/06/2023** – Talk, 93rd Annual Meeting of the International Association of Applied Mathematics and Mechanics (GAMM), Germany.
- **01/06/2023** – Talk, 93rd Annual Meeting of the International Association of Applied Mathematics and Mechanics (GAMM), Germany.
- **05/2023** – Talk, *Orientierungswoche*, Data Science Department, FAU Erlangen–Nürnberg, Germany.
- **09/2022** – Talk, 25th International Symposium on Mathematical Theory of Networks and Systems (MTNS), Germany.
- **30/08/2022** – Talk, *IX Partial Differential Equations, Optimal Design and Numerics*, Spain.
- **24/08/2022** – Talk, *IX Partial Differential Equations, Optimal Design and Numerics*, Spain.
- **04/2022** – Talk, South Zone Mathematics Conference, Chile.

SCIENTIFIC VISITS & ACTIVITIES

- **11/2024** – Participant, Oberwolfach Seminar *Control and Machine Learning*, Mathematisches Forschungsinstitut Oberwolfach, Germany.
- **11/2023** – Scientific visit to Prof. Martin Lazar, University of Dubrovnik, Croatia.
- **10/2023** – Scientific visit to the University of Deusto, Spain, together with Prof. Falk Hante, Humboldt-Universität zu Berlin.
- **10/2020** – Participant, Doctoral School, Pontificia Universidad Católica de Chile, Chile.
- **2019–2020** – Member, Student Council of Sciences, Universidad Técnica Federico Santa María; student representation across five degree programs.
- **01–03/2018** – Student, Summer School, Instituto de Matemática Pura e Aplicada (IMPA), Brazil.

HONORS & AWARDS

- **2026 STAEDTLER Foundation Award** for an outstanding doctoral thesis at FAU.
- Ph.D. awarded *summa cum laude* (highest distinction).
- M.Sc. in Mathematics and Mathematical Engineering degrees awarded with the highest distinction.
- Member of the UTFSM Honor Roll (top 10% by academic performance), 2016–2021.

ADDITIONAL PROFESSIONAL EXPERIENCE

Analyst in Financial Risk 02/2021 – 02/2022

Deloitte SpA, Santiago, Chile

Financial risk modelling and programming in an industry environment.

LANGUAGES & AVAILABILITY

Spanish (native) • English (fluent) • Portuguese (intermediate) • German (basic)